

Rust Memory Management

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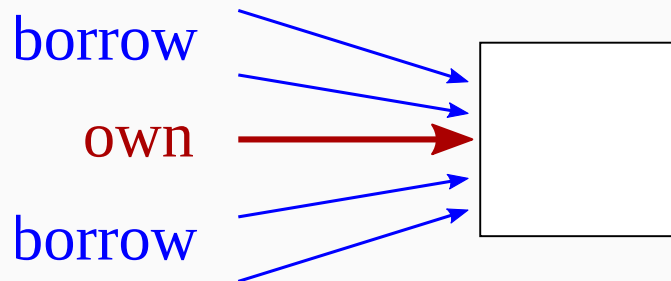
Introduction



Rust's basic idea to memory management

- Rust maintains that, for any live object,
 1. there is *one and only one pointer* that “owns” it (*the owning pointer*)
 2. there are any number of non-owning pointers to it (*borrowing pointers*)
 3. *borrowing pointers cannot be dereferenced after the owning pointer goes away*
- $\Rightarrow$ *it can safely reclaim the data when the owning pointer goes away*

“*single-ownership rule*”



```
{  
  let a = S{x: ..., y: ...};  
  ...  
} // what a points to will be gone here
```

The rules are enforced statically

- Rust enforces the rules (or, detect violations thereof)
 - *statically*, not *dynamically*
 - *compile-time*, not at *runtime*
 - *before* execution, not *during* execution

“borrow checker”

Exception to the single ownership model

- there are actually some exceptions to ways to the rules
 1. **reference counting pointers** ($\approx$ multiple owning pointers)
 - counts the number of owners *at runtime*, and reclaim the data when all these pointers are gone
 2. **unsafe/raw pointers** ($\approx$ totally up to you)

they are not specific to Rust, and we'll not cover them below

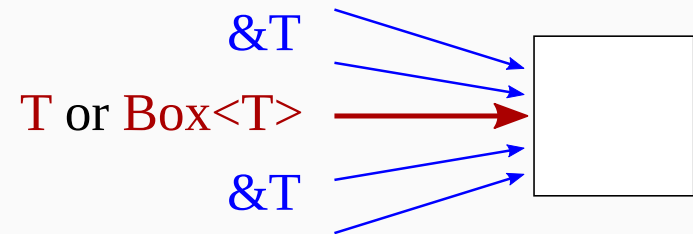
Rust Basics

Pointer-like data types in Rust

given a type T (`i32`, `struct`, `enum`, ...), below are types representing “references (pointers) to T ”

1. T : **owning** pointer to T
2. `Box< $T$ >` (pronounced “*box* T ”) : **owning** pointer to T
3. `& $T$` (pronounced “*ref* T ”) : **borrowing pointer** to T
4. `Rc< $T$ >` and `Arc< $T$ >` : shared (reference-counting) owning pointer to T
5. `* $T$` : unsafe pointer to T

following discussions are focused on
 T , `Box< $T$ >` and `& $T$`



Pointer-making expressions

given an expression e of type T , below are expressions that make pointers to the value of e (besides e itself)

- `Box::new( $e$ )` (of type `Box< $T$ >`) : an owning pointer
- `& $e$` (of type `& $T$`) : a borrowing pointer

An example

```
{  
  let a: S = S{x: ...};           // allocate memory for S  
                                   // and make an owning pointer to it  
  let b: S = a                   // an owning pointer  
  let c: Box<S> = Box::<S>::new(a) // an owning pointer  
  let d: &S = &a                 // a borrowing pointer  
}
```

- note: type of variables can be omitted (spelled out for clarity)
- note: the above program violates several rules so it does not compile

Owning Pointers

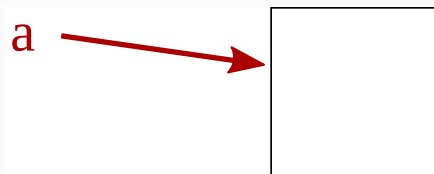
Assignments of owning pointers

- to maintain the “single-owner” rule, an assignment of owning pointers in Rust *does not copy, but moves it* out of the righthand side, disallowing further use of it

`b = a;` *// a cannot be used below*

```
fn foo() {  
    let a = S{x: ..., y: ...};  
    ... a.x ...; // OK, as expected  
    ... a.y ...; // OK, as expected
```

```
}
```

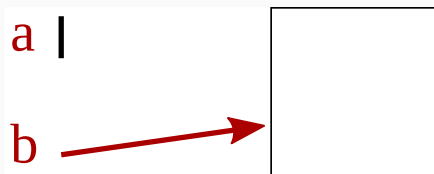


Assignments of owning pointers

- to maintain the “single-owner” rule, an assignment of owning pointers in Rust *does not copy, but moves it* out of the righthand side, disallowing further use of it

```
b = a; // a cannot be used below
```

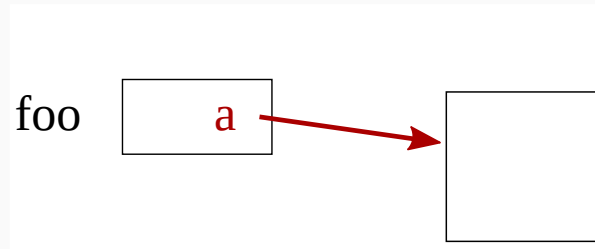
```
fn foo() {  
    let a = S{x: ..., y: ...};  
    ... a.x ...; // OK, as expected  
    ... a.y ...; // OK, as expected  
    // the reference moves out from a  
    let b = a;  
    a.x; // NG, the value has moved out  
    b.x; // OK  
}
```



Argument-passing also moves the reference

- passing a value to a function also moves the reference out of the source

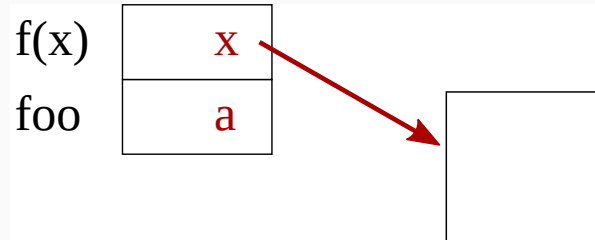
```
fn foo() {  
  let a = S{x: ..., y: ...};  
  ... a.x ...; // OK, as expected  
  ... a.y ...; // OK, as expected  
  
}
```



Argument-passing also moves the reference

- passing a value to a function also moves the reference out of the source

```
fn foo() {  
  let a = S{x: ..., y: ...};  
  ... a.x ...; // OK, as expected  
  ... a.y ...; // OK, as expected  
  // moves the reference out of a  
  f(a);  
  a.x; // NG, the reference has moved  
}
```



Exceptions to “assignment moves the reference”

- you may notice the moving assignment contradicts what you have seen

```
b = a; // a cannot be used after this
```

- if it applies everywhere, does the following program violate it?

```
fn foo() -> f64 {  
    let a = 123.456;  
    let b = a; // does the reference to 123.456 move out from a!?  
    a + 0.789 // if so, is this invalid!?  
}
```

- answer: no, it does *not* apply to primitive types like i32, f64, etc.
- more generally, it does not apply to data types that implement *Copy* trait

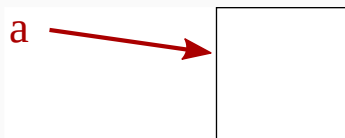
Copy trait

- define your struct with `#[derive(Copy, Clone)]` like

```
#[derive(Copy, Clone)]  
struct S { ... }
```

- $\Rightarrow$ assignment or argument-passing of *S* copies the righthand side

```
fn foo() {  
  let a = S{x: ..., y: ...};  
  a.x; // OK, as expected  
  a.y; // OK, as expected
```



```
}
```

- note: copy types trivially maintain the single-owner rule

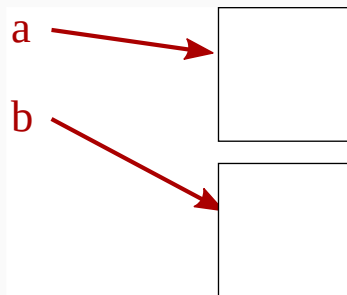
Copy trait

- define your struct with `#[derive(Copy, Clone)]` like

```
#[derive(Copy, Clone)]  
struct S { ... }
```

- $\Rightarrow$ assignment or argument-passing of *S* copies the righthand side

```
fn foo() {  
    let a = S{x: ..., y: ...};  
    a.x; // OK, as expected  
    a.y; // OK, as expected  
    // the value is copied  
    let b = a;  
    a.x; // OK  
    b.x; // OK, too  
}
```



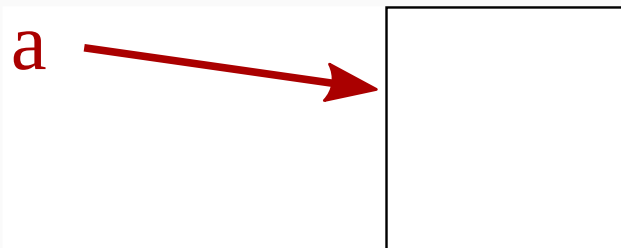
- note: copy types trivially maintain the single-owner rule

Box<*T*> type

Box<T> makes an owning pointer

- making a pointer by `Box::new(v)` moves the reference out of v , too, and `Box::new(v)` becomes the owning pointer

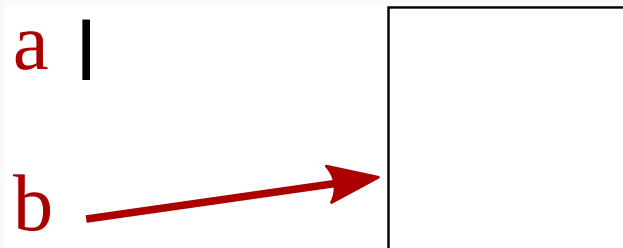
```
fn foo() {  
  let a = S{x: ..., y: ...};  
  a.x; // OK, as expected  
  a.y; // OK, as expected  
}
```



Box<T> makes an owning pointer

- making a pointer by `Box::new(v)` moves the reference out of `v`, too, and `Box::new(v)` becomes the owning pointer

```
fn foo() {  
  let a = S{x: ..., y: ...};  
  a.x; // OK, as expected  
  a.y; // OK, as expected  
  // OK, now b is the owning pointer  
  let b = Box::new(a)  
  a.x; // NG, the value has moved out  
  (*b).x; // OK  
  b.x; // OK. abbreviation of (*b).x  
}
```



Difference between T and `Box<T>`?

- as you have seen, the effects of

```
let b = a
```

and

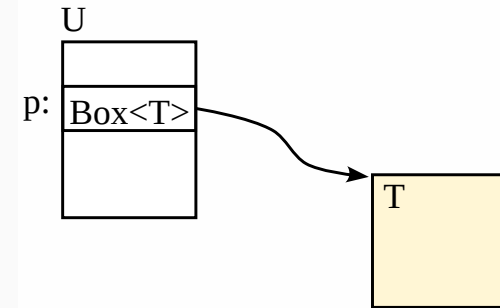
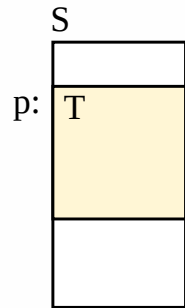
```
let b = Box::new(a)
```

look very similar (identical)

- as far as data lifetime is concerned, it is in fact safe to say they are
- Rust has distinction between them for
 1. specifying data layout
 2. allowing dynamic dispatch only for `Box<T>`
 3. specifying where data are allocated (stack vs. heap)

Data layout differences between T and $\text{Box}\langle T \rangle$

- S and U below have different data layouts
 - struct S { ..., p: T , } “embeds” a T into S
 - struct U { ..., p: $\text{Box}\langle T \rangle$, } has p point to a separately allocated T

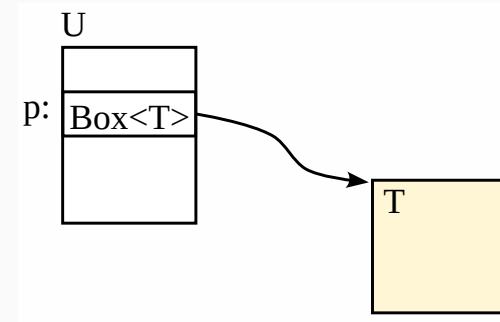
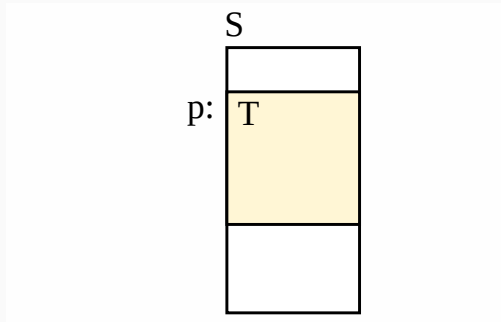


Data layout differences between T and $\text{Box}\langle T \rangle$

- in particular, $\text{Box}\langle T \rangle$ is essential to define recursive data structures
 - `struct S { ..., p: S, }` is not allowed, whereas
 - `struct U { ..., p: Box<U>, }` is
- note: U above can never be constructed; a recursive data structure typically uses `enum` or `Option<Box<..>>`
 - `struct U { ..., p: Option<Box<U>>, }`

Data layout differences between T and $\text{Box}<T>$

- the distinction is insignificant when discussing lifetimes



- in both cases, data of T (yellow box) is gone exactly when the enclosing structure is gone
- another difference is that Rust allocates T on stack and move it to heap when $\text{Box}<T>$ is made
 - but again, it has nothing to do with lifetime (unlike C/C++)

Owning pointers and control flows

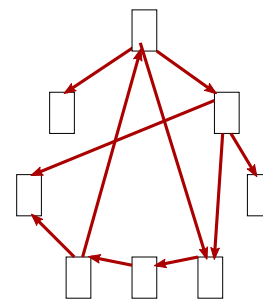
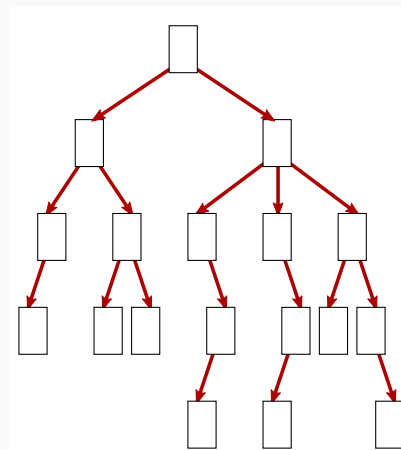
- Rust compiler determines, for each variable of owning pointer type (T or $\text{Box}<T>$), at which point the variable can be *used* (i.e., the value has not been moved out)
- it may be a *conservative* estimate

```
fn foo() {  
    let a = S{x: ..., y: ...};  
    if ... {  
        let b = a;  
    }  
    ... a.x ... // NG  
}
```

```
fn foo() {  
    let a = S{x: ..., y: ...};  
    for ... {  
        let b = a; // NG  
    }  
}
```

A (huge) implication of the single-owner rule

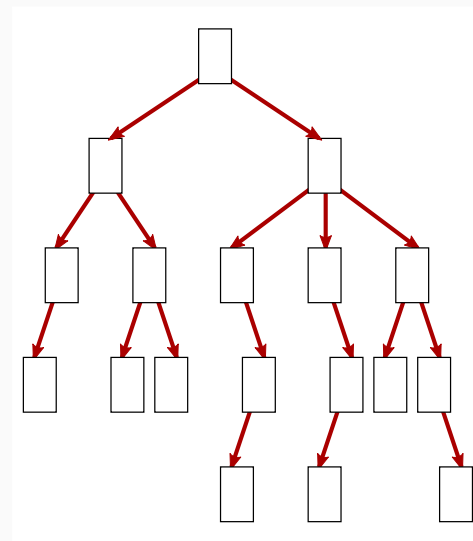
- with only owning pointers (T and $\text{Box}\langle T \rangle$),
 - you can make *a tree* of data,
 - but you *cannot make a general graph* with joins or cycles, where *a node may be pointed to by multiple nodes*
- to make a graph whose nodes are T , use either
 - $\&T$ to represent edges, or
 - $\text{Vec}\langle T \rangle$ to represent nodes and $\text{Vec}\langle (\text{i32}, \text{i32}) \rangle$ to represent edges



The (huge) implication to memory management

- with only owning pointers (i.e., no borrowing pointers)
- *whenever an owning pointer is gone* (e.g.,
 - a variable goes out of scope or
 - a variable or field is overwritten),

the entire tree rooted from the pointer can be safely reclaimed



```
{  
  let t = make_tree(...);  
  ...  
} ... // t deallocated here
```

The (huge) implication to memory management

- Rust exactly does that, with the additional guarantee that *borrowing pointers are never dereferenced after its owning pointer is gone*

Motto:

lifetime of data = lifetime of its owning pointer
= program points its owning pointer can be dereferenced ($\dagger$)
 $\approx$ the block its owning pointer variable is defined

```
{  
  let s = S{ ... }; // or Box::new(S{...})  
  ...  
  ...  
} ... // referent of s reclaimed here
```

- ($\dagger$) : determined by control flows and assignments, to be precise

Borrowing pointers ($\&T$)

- you can derive any number of borrowing pointers ($\&T$) from T or $\text{Box}\langle T \rangle$
- the owning pointer remains valid after a borrowing pointer has been made

```
let a = S{x: .., y: ..};  
let b = &a;  
... a.x + b.x ... // OK
```

- the issue is how to prevent a program from *dereferencing borrowing pointers after its owning pointer is gone*

Borrowers rule in action

- a borrowing pointer cannot be dereferenced after its owning pointer is gone

```
fn foo() -> i32 {  
    let c: &S; // a reference to S
```

c : &S

```
}
```

Borrowers rule in action

- a borrowing pointer cannot be dereferenced after its owning pointer is gone

```
fn foo() -> i32 {  
    let c: &S; // a reference to S  
    { // an inner block  
        let b: &S; // another reference  
  
    }  
}
```

c : &S

b : &S

Borrowers rule in action

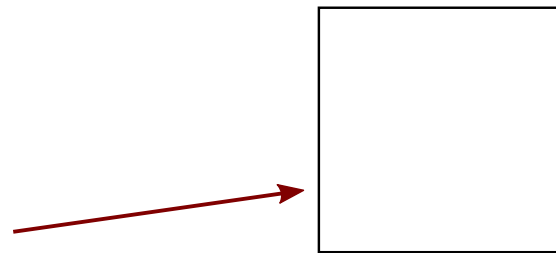
- a borrowing pointer cannot be dereferenced after its owning pointer is gone

```
fn foo() -> i32 {  
  let c: &S; // a reference to S  
  { // an inner block  
    let b: &S; // another reference  
    let a = S{x: ...}; // allocate S  
  
  }  
}
```

c : &S

b : &S

a : S



Borrowers rule in action

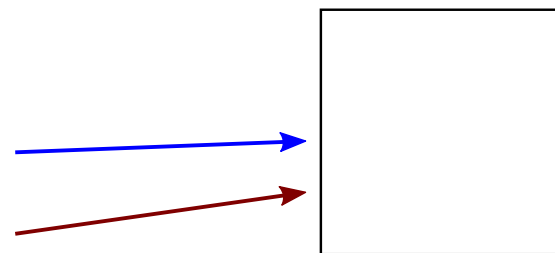
- a borrowing pointer cannot be dereferenced after its owning pointer is gone

```
fn foo() -> i32 {  
  let c: &S; // a reference to S  
  { // an inner block  
    let b: &S; // another reference  
    let a = S{x: ...}; // allocate S  
    // OK (both a and b live only until the end of  
    // the inner block)  
    b = &a;  
  }  
}
```

c : &S

b : &S

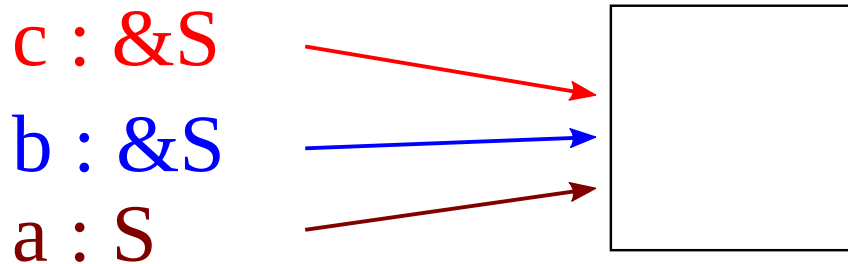
a : S



Borrowers rule in action

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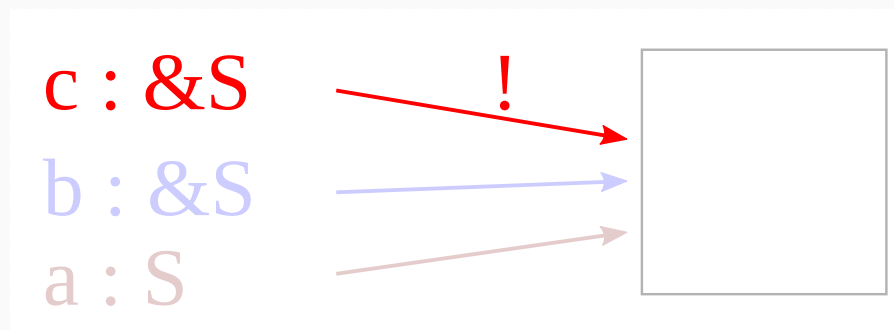
```
fn foo() -> i32 {  
  let c: &S; // a reference to S  
  { // an inner block  
    let b: &S; // another reference  
    let a = S{x: ...}; // allocate S  
    // OK (both a and b live only until the end of  
    // the inner block)  
    b = &a;  
    c = b; // dangerous (c outlives a)  
  }  
}
```



Borrowers rule in action

- a borrowing pointer cannot be dereferenced after its owning pointer is gone

```
fn foo() -> i32 {  
  let c: &S; // a reference to S  
  { // an inner block  
    let b: &S; // another reference  
    let a = S{x: ...}; // allocate S  
    // OK (both a and b live only until the end of  
    // the inner block)  
    b = &a;  
    c = b; // dangerous (c outlives a)  
  } // a dies here, making c a dangling pointer  
  c.x // NG (deref a dangling pointer)  
}
```



A *mutable* borrowing reference (&mut T)

- data cannot be modified through ordinary borrowing references &T

```
let a : S = S{x: 10, y: 20};  
let b : &S = &a;  
b.x = 100; // NG
```

- i.e., &T is the type of *immutable* references
- you can modify data only through *a mutable reference* (&mut T)

```
let mut a : S = S{x: 10, y: 20};  
let b : &mut S = &mut a;  
b.x = 100; // OK
```

- the difference is largely orthogonal to memory management

Borrow-checking details

A technical remark about the borrow-checking

- it's *not* dangling pointers, *per se*, that are prevented, but their *dereferencing*
- *the previous code compiles* as long as *c* is not dereferenced

```
fn foo() -> i32 {  
  let c: &S; // a reference to S  
  { // an inner block  
    let b: &S; // another reference  
    let a = S{x: ...}; // allocate S  
    // OK (both a and b live only until the end of the inner block)  
    b = &a;  
    c = b; // dangerous (c outlives a)  
  } // a dies here, making c a dangling pointer  
  // c.x don't deref c  
}
```

How borrow-checking works : lifetime

- *lifetime* of data
 - = program points where the data has not been deallocated
 - = program points where the data's owning pointer is valid
- for each borrowing pointer, Rust compiler determines the *lifetime* of data it points to (*referent lifetime*) as its static type
- upon assignment $p = q$ between borrowing pointers, it demands
referent lifetime of $p \subset$ referent lifetime of q

How borrow-checking basically works

```
fn foo() -> i32 {  
    let c: &S;  
    {  
        let b: &S;  
        let a = S{x: ...}; // lives until  $\alpha$   
  
    }  
  
}
```

1. the owning pointer a's lifetime is the inner block; call it α (...)
2. let β and γ be referent lifetimes of b and c, respectively

How borrow-checking basically works

```
fn foo() -> i32 {  
  let c: &S;  
  {  
    let b: &S;  
    let a = S{x: ...}; // lives until  $\alpha$   
      
b = &a; // b's referent lifetime  $\subset$  a's =  $\alpha$   
c = b; // c's referent lifetime  $\subset$  b's =  $\alpha$   
 ...  $\alpha$   
  }  
}
```

1. the owning pointer a's lifetime is the inner block; call it α (...)
2. let β and γ be referent lifetimes of b and c, respectively
3. due to the assignments,
 - $b = \&a \Rightarrow \beta \subset \alpha$
 - $c = b \Rightarrow \gamma \subset \beta (\subset \alpha)$

How borrow-checking basically works

```
fn foo() -> i32 {  
    let c: &S;  
    {  
        let b: &S;  
        let a = S{x: ...}; // lives until  $\alpha$   
          
b = &a; // b's referent lifetime  $\subset$  a's =  $\alpha$   
c = b; // c's referent lifetime  $\subset$  b's =  $\alpha$   
         ...  $\alpha$   
    }  
    c.x // NG (deref outside c's referent lifetime =  $\alpha$ )  
}
```

1. the owning pointer a's lifetime is the inner block; call it α ()
2. let β and γ be referent lifetimes of b and c, respectively
3. due to the assignments,
 - $b = \&a \Rightarrow \beta \subset \alpha$
 - $c = b \Rightarrow \gamma \subset \beta (\subset \alpha)$
4. dereference c.x must be $\subset \gamma (\subset \alpha)$, which is not the case (i.e., invalid)

Programming with borrowing references

- in more general cases, programs using borrowing references must help compilers track their referent lifetimes
- this must be done for functions called from unknown places, function calls to unknown functions and data structures
- to this end, the programmer sometimes must annotate *reference types with their referent lifetimes*

References in function calls

- how to check the validity of a functions call without knowing its body?

```
{
  let r : &i32;
  let a = 123;
  {
    {
      let b = 456;
      {
        let c = 789;
        r = foo(&a, &b, &c); ...  $\gamma$ 
      } ...  $\beta$ 
    }
  } ...  $\alpha$ 
  *r // ( $\dagger$ )
}
```

- ▶ $*r$ should be safe if $f(p, q, r)$ returns a reference whose referent lifetime contains ($\dagger$); i.e., p

References in data structures

- how to check the validity of dereferencing references obtained from a data structure

```
struct A { b : &B }  
struct B { c : &C }  
struct C { x : i32 }  
  
...  
let c = C{x : 123};  
let b = B{c : &c};  
let mut a = A{b : &b};
```

`a.b.c.x` // OK?

References in data structures

- how to check the validity of dereferencing references obtained from a data structure

```
struct A { b : &B }  
struct B { c : &C }  
struct C { x : i32 }  
  
...  
let c = C{x : 123};  
let b = B{c : &c};  
let mut a = A{b : &b};  
{  
    let b2 = B{c : &c};  
    a.b = &b2;  
}  
a.b.c.x // OK?
```

References in function parameters

- how to check the validity of functions taking references or structures containing references, *without knowing all its callers*

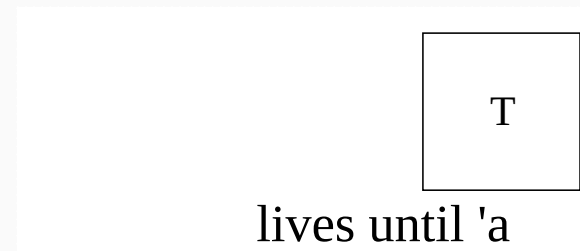
```
fn bar(a : &mut i32, b : &i32) {  
    *a = b;  
}
```

- what if references are in structures ...

```
fn baz(a : &mut A, b: &B) {  
    a.b = b  
}
```

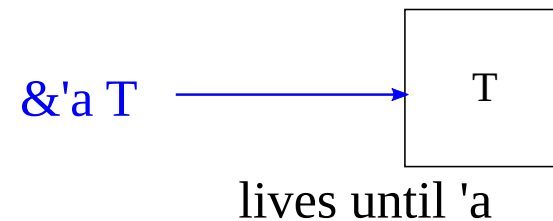
Reference type with a lifetime parameter

- to address these problems, Rust's borrowing reference types ($\&T$ or $\&\text{mut } T$) carry *lifetime parameter representing their referent lifetimes*



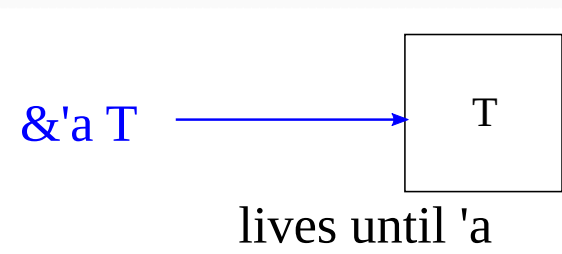
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- syntax:
 - $\&'a T$: reference to “ T whose lifetime is ‘ a ”
 - $\&'a \text{mut } T$: ditto; except you can modify data through it



Reference type with a lifetime parameter

- to address these problems, Rust's borrowing reference types ($\&T$ or $\&\text{mut } T$) carry *lifetime parameter representing their referent lifetimes*
- syntax:
 - $\&'a T$: reference to “ T whose lifetime is ‘ a ”
 - $\&'a \text{mut } T$: ditto; except you can modify data through it
- *every* reference carries a lifetime parameter, though there are places you can omit them
- roughly, you must write them explicitly in function parameters, return types, and struct/enum fields; and can omit them for local variables



Attaching lifetime parameters

- rule: reference types that appear in function parameters, return types, and struct/enum fields must have explicit lifetime parameters

Attaching lifetime parameters to functions

- therefore the following does not compile:

```
fn foo(ra: &i32, rb: &i32, rc: &i32) -> &i32 {  
    ra  
}
```

with errors like:

```
|  
| fn foo(ra: &i32, rb: &i32, rc: &i32) -> &i32 {  
|          ----      ----      ----      ^ expected named lifetime parameter  
|
```

= help: this function's return type contains a borrowed value, but the signature does not say whether it is borrowed from `ra`, `rb`, or `rc`

help: consider introducing a named lifetime parameter

```
|  
| fn foo<'a>(ra: &'a i32, rb: &'a i32, rc: &'a i32) -> &'a i32 {  
|          ++++      ++          ++          ++          ++  
|
```

Why do we need an annotation, *fundamentally*?

- without any annotation, how to know whether this is safe, *without knowing the body of foo?*

```
{
  let r : &i32;
  let a = 123;
  {
    let b = 456;
    {
      let c = 789;
      r = foo(&a, &b, &c);
    }
  }
  *r
}
```

- essentially, the compiler complains “tell me what kind of referent lifetime the reference returned by `foo(&a, &b, &c)` has”
- it must be inferred without knowing the body of `foo`, only from its type

Attaching lifetime parameters

- functions

```
fn f<'a, 'b, 'c, ...> (p0 : T0, p1 : T1, ...) -> Tr { ... }
```

- structs/enums

```
struct A<'a, 'b, 'c, ...> {  
    f0 : T0,  
    f1 : T1,  
    ...  
}
```

- $T_0, T_1, \dots$, and T_r may use 'a, 'b, 'c, ... as lifetime parameters (e.g., &'a i32)

One way to attach lifetime parameters to the example

```
fn foo<'a>(ra: &'a i32, rb: &'a i32, rc: &'a i32) -> &'a i32
```

- effect: the return value is assumed to point to the shortest of the three
- why? generally, when Rust compiler finds `foo(x, y, z)`, it tries to determine 'a so that 'a $\subset$ referent lifetimes of x, y, and z
- in this case,

'a $\subset$ (life time of a) $\cap$ (life time of b) $\cap$ (life time of c) = life time of c

- as a result, our program does not compile, even if `foo(&a, &b, &c)` in fact returns &a

```
{
  let r: &i32;
  let a = 123;
  {
    let b = 456;
    {
      let c = 789;
      r = foo(&a, &b, &c);
      // 'a  $\leftarrow \alpha \cap \beta \cap \gamma = \gamma$ 
      // and r's type becomes & $\gamma$  i32
    } // c's lifetime (=  $\gamma$ ) ends here
  } // b's lifetime (=  $\beta$ ) ends here
  *r // NG, as we are outside  $\gamma$ 
} // a's lifetime (=  $\alpha$ ) ends here
```

An annotation that works

```
fn foo<'a, 'b, 'c>(ra: &'a i32, rb: &'b i32, rc: &'c i32) -> &'a i32
```

- signifies that the return value points to data whose lifetime is `ra`'s referent lifetime (and has nothing to do with `rb`'s or `rc`'s)
- for `foo(x, y, z)`, Rust compiler tries to determine `'a` so that `'a  $\subset$  referent lifetimes of x`
- as a result, the program we are discussing compiles

```
{  
  let r: &i32;  
  let a = 123;  
  {  
    let b = 456;  
    {  
      let c = 789;  
      r = foo(&a, &b, &c);  
      // 'a  $\leftarrow \alpha$   
      // and r's type becomes  $\&\alpha$  i32  
    } // c's lifetime (=  $\gamma$ ) ends here  
  } // b's lifetime (=  $\beta$ ) ends here  
  *r // OK, as here is within  $\alpha$   
} // a's lifetime (=  $\alpha$ ) ends here
```

Types with lifetime parameters capture/constrain the function's behavior

- what if you try to fool the compiler by:

```
fn foo<'a, 'b, 'c>(ra: &'a i32, rb: &'b i32, rc: &'c i32) -> &'a i32 {  
    rb  
}
```

- the compiler rejects returning `rb` (of type `&'b`) when the function's return type is `&'a`, as it cannot infer

lifetime represented by `'a` $\subset$ lifetime represented by `'b`

References in data structures

```
struct A { b : &B }  
struct B { c : &C }  
struct C { x : i32 }  
  
fn baz(a : &mut A, b: &B) {  
    a.b = b  
}
```

$\Rightarrow$

```
struct C { x : i32 }
```

does not compile

References in data structures

```
struct A { b : &B }  
struct B { c : &C }  
struct C { x : i32 }  
  
fn baz(a : &mut A, b: &B) {  
    a.b = b  
}
```

$\Rightarrow$

```
struct B<'c> { c : &'c C }  
struct C { x : i32 }
```

does not compile

References in data structures

```
struct A { b : &B }  
struct B { c : &C }  
struct C { x : i32 }  
  
fn baz(a : &mut A, b: &B) {  
    a.b = b  
}
```

⇒

```
struct A<'b, 'c> { b : &'b B<'c> }  
struct B<'c> { c : &'c C }  
struct C { x : i32 }
```

does not compile

References in data structures

```
struct A { b : &B }  
struct B { c : &C }  
struct C { x : i32 }  
  
fn baz(a : &mut A, b: &B) {  
    a.b = b  
}
```

does not compile

⇒

```
struct A<'b,'c> { b : &'b B<'c> }  
struct B<'c> { c : &'c C }  
struct C { x : i32 }  
  
fn baz<'a,'b,'c','d','e> (a : &'a mut A<'b,'c>,  
                           b: &'d B<'e>) {  
    a.b = b  
}
```


References in data structures

```
struct A { b : &B }  
struct B { c : &C }  
struct C { x : i32 }  
  
fn baz(a : &mut A, b: &B) {  
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struct A<'b,'c> { b : &'b B<'c> }  
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fn baz<'a,'b,'c','d','e> (a : &'a mut A<'b,'c>,  
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    a.b = b  
}
```

does not compile

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struct A { b : &B }  
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struct A<'b,'c> { b : &'b B<'c> }  
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struct C { x : i32 }
```

```
fn baz<'a,'b,'c>(a : &'a mut A<'b,'c>,  
                 b: &'b B<'c>) {  
    a.b = b  
}
```

References in data structures

```
struct A { b : &B }  
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struct C { x : i32 }  
  
fn baz(a : &mut A, b: &B) {  
    a.b = b  
}
```

does not compile

⇒

```
struct A<'b,'c> { b : &'b B<'c> }  
struct B<'c> { c : &'c C }  
struct C { x : i32 }  
  
fn baz<'a,'b,'c>(a : &'a mut A<'b,'c>,  
                 b: &'b B<'c>) {  
    a.b = b  
}
```

does compile

Dereferencing data structure

- as stated earlier, dereferencing a borrowing pointer of type $\&'a \dots$ is allowed at program point p when:

$$p \subset \text{lifetime represented by } 'a$$

- the rule is actually more strict; for types involving lifetime parameters (e.g., $A<'a, 'b, 'c, \dots>$), the above applies to *all* parameters

Dereferencing data structure

- the following program is *safe*, but rejected by the compiler

```
struct S<'a,'b> {  
  a : &'a i32,  
  b : &'b i32,  
}  
...  
let a = 123;  
let mut s = S{a: &a, b: &a};  
{  
  let b = 456;  
  s.b = &b;  
}  
// s.b is a dangling pointer, but s.a is not  
*s.a ... (†)
```

Dereferencing data structure

- the following program is *safe*, but rejected by the compiler

```
struct S<'a,'b> {  
  a : &'a i32,  
  b : &'b i32,  
}  
...  
let a = 123;  
let mut s = S{a: &a, b: &a};
```

```
{  
  let b = 456;  
  s.b = &b;  
}
```

 ... β

// s.b is a dangling pointer, but s.a is not

**s.a ... ($\dagger$)*

```
error[E0597]: `b` does not live long enough  
--> str.rs:11:15  
10 |         let b = 456;  
    |         - binding `b` declared here  
11 |         s.b = &b;  
    |         ^^ borrowed value does not live long enough  
12 |     }  
    |     - `b` dropped here while still borrowed  
13 |     *s.a  
    |     ---- borrow later used here
```

Dereferencing data structure

- the following program is *safe*, but rejected by the compiler

```
struct S<'a,'b> {  
  a : &'a i32,  
  b : &'b i32,  
}  
...  
let a = 123;  
let mut s = S{a: &a, b: &a};
```

```
{  
  let b = 456;  
  s.b = &b;  
}
```

 ... β

// s.b is a dangling pointer, but s.a is not

**s.a ... ($\dagger$)*

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12 |     }  
    |     - `b` dropped here while still borrowed  
13 |     *s.a  
    |     ---- borrow later used here
```

- $s.a$ is not allowed, because:
 - the type of s is $S<'a, 'b>$ and
 - $'b \subset \beta$ ($\because s.b = \&b$);
 - $\therefore \dagger \notin 'b$

Lifetime parameters in a function

- because of this restriction, the compiler can assume all lifetime parameters that appear in the function parameters contain the function body
- the compiler deduces dereferencing `a.b` below is safe based on this assumption

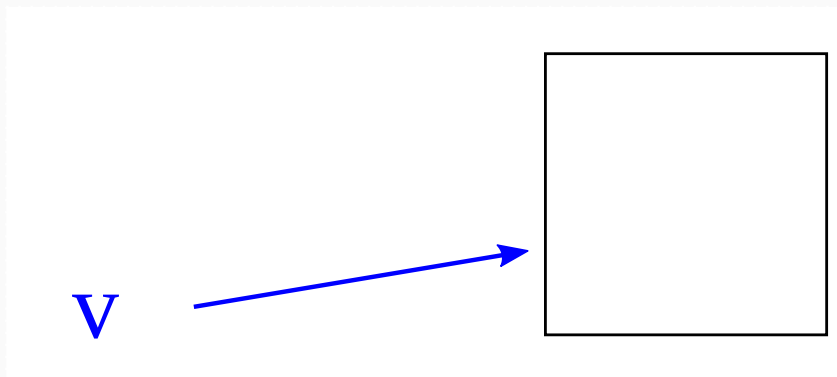
```
fn baz<'a, 'b, 'c'>(a : &'a mut A<'b, 'c>,
                  b : &'b B<'c>) {
    a.b = b
}
```


Summary

Why memory management is difficult

- every language wants to prevent *dereferencing a pointer to an already-reclaimed memory block (dangling pointer)*
- the problem would have been trivial if *you could reclaim v 's referent as soon as v goes out of scope*
- this is not the case, as *v 's referent may still be reachable from other variables when v goes out of scope*

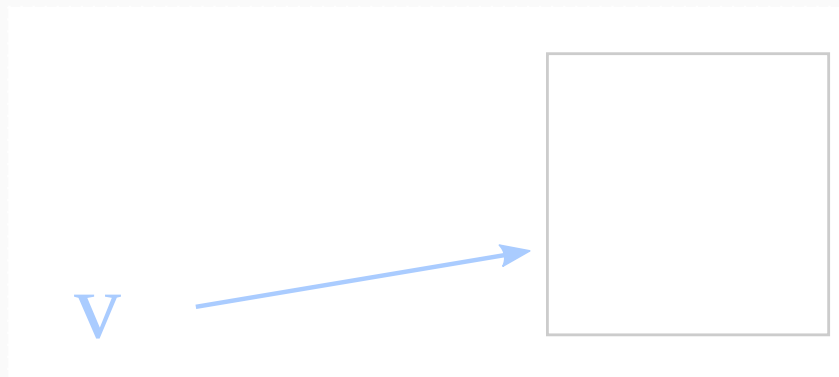
```
let p : &T;  
{  
  let v = T{x: ...};  
  ...  
  p = &v;  
} // v never used below, but its referent is  
... p.x ...
```



Why memory management is difficult

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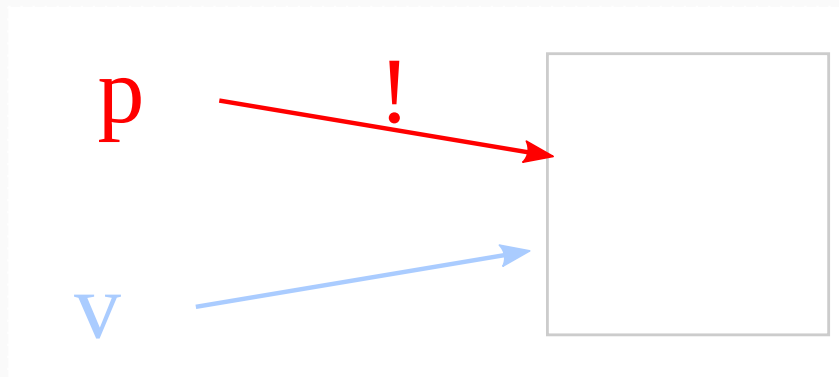
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{  
  let v = T{x: ...};  
  ...  
  p = &v;  
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```



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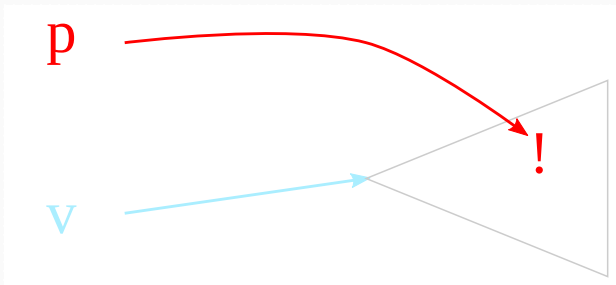
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```



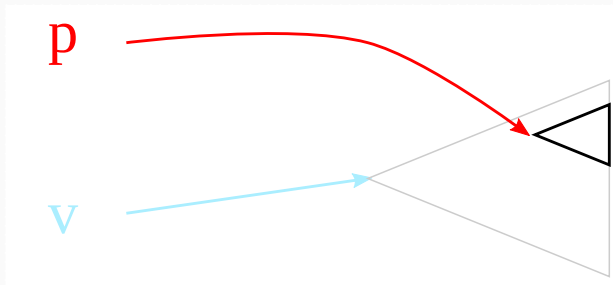
C vs. GC vs. Rust

- C/C++ : it's up to you
- GC : if it is reachable from other variables, I retain it for you
- Rust : when v goes out of scope,
 1. I reclaim T_v , all data *reachable from v through owning pointers*
 2. T_v may be reachable from other variables via borrowing references, but I guarantee such references are never dereferenced

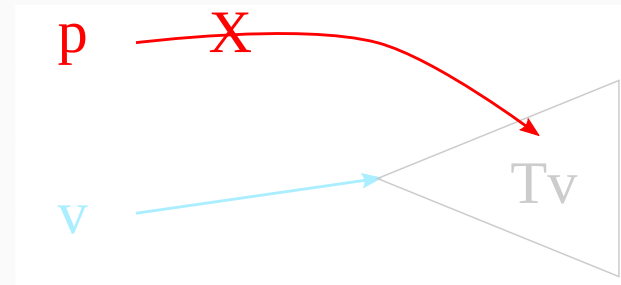
C/C++



GC

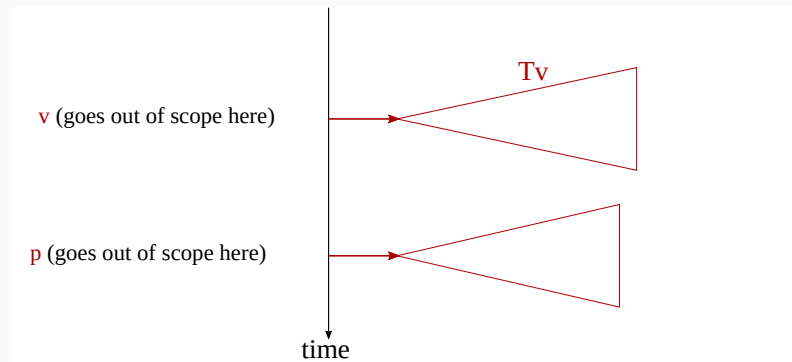


Rust



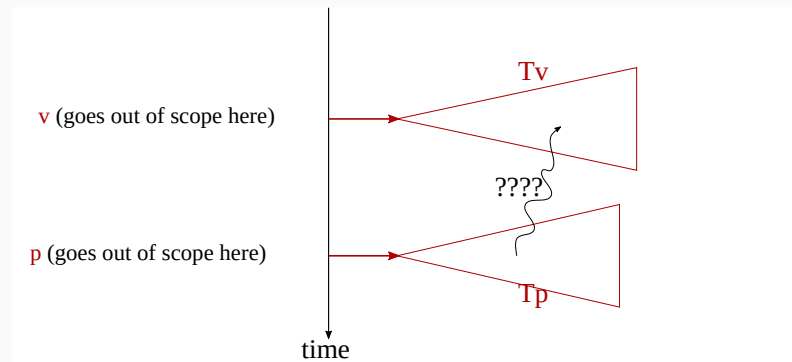
How Rust achieves it?

- say two data structures T_v rooted at variable v and T_p rooted at variable p
- assume v goes out of scope earlier than p
- we wish to guarantee when v goes out of scope, it is safe to reclaim the entire T_v
- generally it is of course not the case, as there may be pointers somewhere in $T_p \rightarrow$ somewhere in T_v



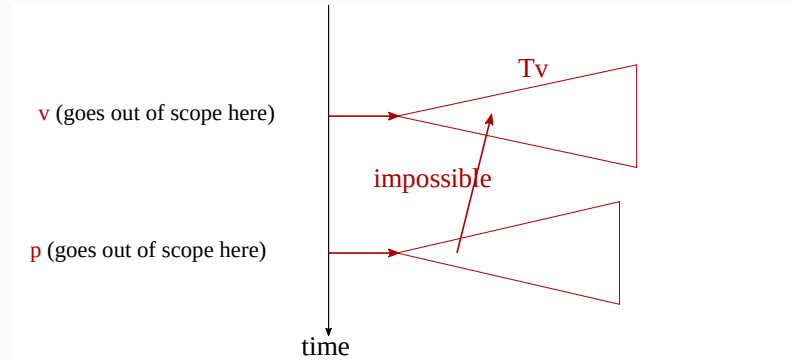
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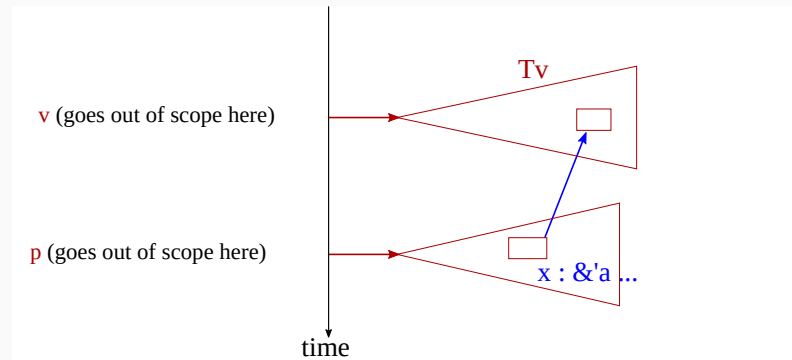
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- recall the “single-owner rule,” which guarantees there is only one owning pointer to any node
- $\Rightarrow$ there can be *no owning pointers* from outside T_v to inside T_v



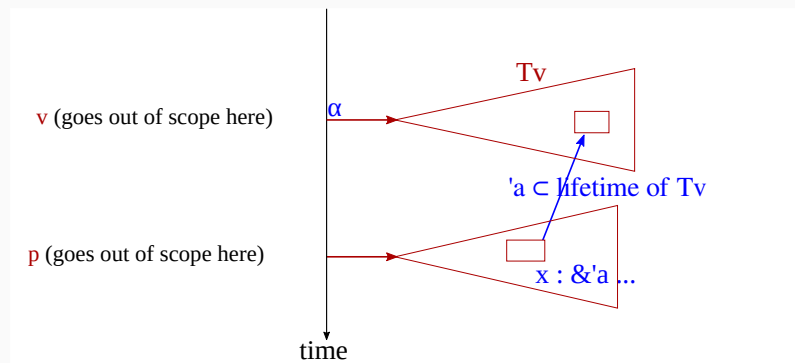
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- $\Rightarrow$ any such pointer must be *a borrowing pointer*



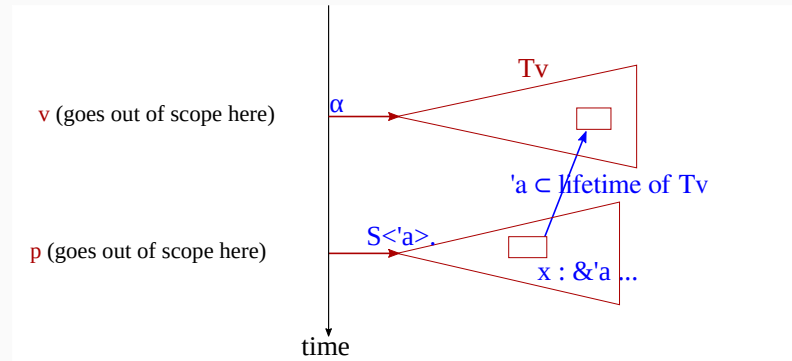
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- $\Rightarrow$ there can be *no owning pointers* from outside T_v to inside T_v
- $\Rightarrow$ any such pointer must be *a borrowing pointer*
- recall that a borrowing pointer must have a lifetime parameter; e.g., `'a`
- it must hold that `'a` $\subset$ lifetime of T_v



How Rust achieves it?

- any structure containing borrowing pointers must have these parameters as part of its type (e.g., `S<'a>`)
- by `'a  $\subset$  lifetime of  $T_v$` , the containing data structure (of type `S<'a>`) cannot be dereferenced after T_v is gone



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- by `'a ⊂ lifetime of  $T_v$` , the containing data structure (of type `S<'a>`) cannot be dereferenced after T_v is gone

